

The empirical recurrence for OEIS A281931, proved by counting patterns with a budget

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Abstract

OEIS A281931 imposes three things at once: a condition on each cell that reaches one row up and one row down; a global budget allowing exactly two cells to break it; and the clause “new values introduced in order 0 sequentially upwards”, which says the array is written in canonical form. None of the three is visible to a plain transfer matrix over rows. All three are handled: the state is a pair of consecutive rows together with the number of offending cells so far, capped at the budget; and the canonical-form clause is removed by counting equality patterns instead of arrays, through the identity $2!a(n) = \sum_i \binom{2}{i} D_{2-i} L_i(n)$ with D_m the derangement numbers. The empirical recurrence of order 21 contributed by R. H. Hardin then follows from a finite exact computation on $S = 195$ states.

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1 The sequence and the conjecture

OEIS A281931 is “Number of $n \times 3$ 0..1 arrays with no element equal to more than four of its king-move neighbors, with the exception of exactly two elements, and with new values introduced in order 0 sequentially upwards.”. It has offset 1 and begins

0, 1, 20, 312, 3573, 31410, 252630, 1925590, ...

The entry carries the following comment:

Empirical: $a(n) = 15*a(n-1) - 63*a(n-2) + 23*a(n-3) + 3*a(n-4) + 1161*a(n-5) + 208*a(n-6) - 2013*a(n-7) - 8424*a(n-8) - 6999*a(n-9) + 7797*a(n-10) + 25689*a(n-11) + 44356*a(n-12) + 27282*a(n-13) - 498*a(n-14) - 60721*a(n-15) - 115722*a(n-16) - 113223*a(n-17) - 120471*a(n-18) - 77700*a(n-19) - 35280*a(n-20) - 21952*a(n-21)$ for $n > 26$

As of the “Last modified” line on the live entry (Feb 02 2017 08:19:36, revision 4) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) = 15a(n-1) - 63a(n-2) + 23a(n-3) + 3a(n-4) + 1161a(n-5) + 208a(n-6) - 2013a(n-7) - 8424a(n-8) - 6999a(n-9) + 7797a(n-10) + 25689a(n-11) + 44356a(n-12) + 27282a(n-13) - 498a(n-14) - 60721a(n-15) - 115722a(n-16) - 113223a(n-17) - 120471a(n-18) - 77700a(n-19) - 35280a(n-20) - 21952a(n-21) \quad (1)$$

for all $n > 26$.

2 The condition, the budget, and the canonical form

The arrays have 3 columns and $L = n$ rows; write $x_{t,u}$ for the entry in row t and column u . With the neighbour offsets

$$\mathcal{N} = \{(0, -1), (0, 1), (-1, 0), (1, 0), (-1, -1), (1, 1), (-1, 1), (1, -1)\}$$

as (row shift, column shift), counting only positions inside the array, call a cell (t, u) *offending* when

$$\#\{\text{neighbours equal to } x_{t,u}\} > 4.$$

The entry asks for arrays in which exactly $E = 2$ cells offend.

Since $\mathcal{N}$ contains offsets of both signs in the row direction, a cell of row t is judged by rows $t-1$, t and $t+1$ together, so a state carrying one row is not enough; and the number of offending cells is a property of the whole array, so no bounded window sees it either. Both are fixed by the same device: states are ordered pairs of consecutive rows together with a running count $v \in \{0, 1, \dots, 2\}$, and a count that would exceed 2 is simply not represented — an array that passes it can never come back under it.

The condition is stated in equalities between entries, so it is invariant under relabelling the values, and so is the number of offending cells. The clause “new values introduced in order 0 sequentially upwards” keeps exactly one array from each relabelling class, so what the entry counts is admissible equality PATTERNS with at most $K = 2$ classes.

Lemma 1. *Let $L_i(n)$ count the arrays over an alphabet of i letters with exactly 2 offending cells, and $N_j(n)$ the admissible patterns with exactly j classes. Then $L_i = \sum_j N_j i(i-1) \cdots (i-j+1)$ and*

$$2! \sum_{j \leq 2} N_j(n) = \sum_{i=0}^2 \binom{2}{i} D_{2-i} L_i(n), \quad D_m = m! \sum_{t=0}^m \frac{(-1)^t}{t!}.$$

Proof. A labelled array is a pattern together with an injection of its classes into the alphabet, and a pattern with j classes admits $i(i-1) \cdots (i-j+1)$ of them, which gives the first identity. Writing the falling factorial as $j! \binom{i}{j}$ and inverting binomially, $j! N_j = \sum_i (-1)^{j-i} \binom{j}{i} L_i$; summing over $j \leq 2$ and exchanging the order of summation makes the coefficient of L_i equal to $\sum_{t \leq 2-i} (-1)^t / (i! t!)$, which multiplied by $2!$ is $\binom{2}{i} D_{2-i}$, an integer. $\square$

3 Each L_i is a walk count

Fix i , let $V_i = \{0, \dots, i-1\}^3$, and take the triples (p, c, v) with $p, c \in V_i$ and $0 \leq v \leq 2$ as vertices. Write $\text{viol}(b \mid a, x)$ for the number of offending cells in the row b when a lies above it and x below (either may be absent, in which case the corresponding neighbours simply do not exist). Declare

$$(p, c, v) \longrightarrow (c, x, v + \text{viol}(c \mid p, x))$$

an edge whenever the new count is at most 2. Let $\mathbf{s}_i$ mark the vertices with $v = \text{viol}(p \mid -, c)$ and $\mathbf{e}_i$ those with $v + \text{viol}(c \mid p, -) = 2$.

Lemma 2. *For $L \geq 2$, $L_i = \mathbf{s}_i^\top P_i^{L-2} \mathbf{e}_i$, where P_i is the adjacency matrix of that digraph.*

Proof. Consecutive vertices overlap in one row, so a walk of length $L-2$ is exactly a sequence of L rows, that is, an array. Along it, the cells of row t for $2 \leq t \leq L-1$ are judged once, at the step leaving $(r^{(t-1)}, r^{(t)}, \cdot)$, with their true neighbours; those of row 1 are judged by $\mathbf{s}_i$ with no row above, and those of row L by $\mathbf{e}_i$ with no row below. Hence the running count is the number of offending cells among the rows judged so far, and the walk exists exactly when that total never exceeds 2; since the total only grows, that is implied by the final value being 2, which $\mathbf{e}_i$ imposes. $\square$

Assembling the 2 chains block-diagonally into N of size $S = 195$ with the weights of Lemma 1 in the start vector $\mathbf{v}$ and the end markers in $\mathbf{u}$,

$$2! a(n) = \mathbf{v}^\top N^{L-2} \mathbf{u}, \quad L = n. \quad (2)$$

4 The criterion, and the computation

Let $q(t) = t^{21} - \sum_i c_i t^{21-i}$ be the characteristic polynomial of (1) and $G = N$.

Lemma 3. *With n_0 the least index for which $L \geq 2$, $o = 1n_0 + 0 - 2$ and $h_j = G^j N^o \mathbf{u}$, we have for $j \geq 21$*

$$2! \left(a(n_0 + j) - \sum_i c_i a(n_0 + j - i) \right) = \mathbf{v}^\top G^{j-21} w, \quad w = h_{21} - \sum_i c_i h_{21-i}.$$

Hence (1) holds from that point on if and only if $\mathbf{v}^\top G^m w = 0$ for all $m \geq 0$, and $m < S$ suffices.

Proof. Substitute (2) and factor; $2!$ is a nonzero scalar. Cayley–Hamilton makes G^S an integer combination of $I, G, \dots, G^{S-1}$, so every later value repeats that combination of earlier ones. $\square$

Theorem 4. *The recurrence (1) holds for every $n > 26$, which is the statement of Section 1.*

Proof. The digraph was built directly from the definitions above. The vector w was formed by 21 applications of G in exact integer arithmetic and $\mathbf{v}^\top G^m w$ evaluated for $m = 0, 1, \dots$, again exactly. Every one of those integers vanishes from the index corresponding to $n = 26 + 1$ onwards, and the run continued until $S + 1$ consecutive zeros had been seen. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the right-hand side of (2), divided by $2!$, was evaluated at every one of the 20 published indices and agrees with the entry’s DATA exactly, the quotient coming out an integer each time; no shift was fitted, the number of rows being $L = n$ from the entry’s own name and the offset saying which index the first published term carries. A misreading of the offending-cell rule, of the budget, or of the inversion in Lemma 1 would change the counts and be rejected here.

Second, the recurrence (1) was evaluated on those published terms in exact integer arithmetic, with no matrices involved, and vanishes wherever it applies.

Third, the annihilation test of Lemma 3 was carried to the full Cayley–Hamilton bound in exact integer arithmetic rather than to a sampled prefix.

References

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